

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	(a)	(i)		Motion of molecules remains unchanged (1) Collisions occur more often (1) Don't accept more collisions	2			2		2
		(ii)		It / volume decreases (1) to half or 9 cm ³ (1) Award 2 marks for inversely proportional so it halves		2		2	2	2
	(b)	(i)		300 [K]		1		1	1	1
		(ii)		Substitution: $\frac{2.4 \times 10^5}{300 \text{ ecf}} = \frac{3.0 \times 10^5}{T_2}$ (1) in any correct arrangement $T_2 = \frac{300 \text{ ecf} \times 3.0 \times 10^5}{2.4 \times 10^5} = 375 \text{ [K]} (1) (= 102^\circ\text{C})$ Temp rise = $375 - 300 \text{ ecf} = 75 \text{ [K]} \text{ or } 75^\circ\text{C} (1)$ Alternative: $\frac{p_1}{p_2} = \frac{2.4 \times 10^5}{3.0 \times 10^5} = 0.8 (1)$ $\frac{T_1}{T_2} = 0.8 \text{ so } T_2 = \frac{300 \text{ ecf}}{0.8} = 375 \text{ [K]} (1)$ Temp rise = $375 - 300 \text{ ecf} = 75 \text{ [K]} \text{ or } 75^\circ\text{C} (1)$	1	1 1		3	3	3